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1. The dynamics and the observations

We assume a deterministic dynamical system with a state z_t that evolves according to a time-independent dynamics Φ ,

$$z_{t+1} = \Phi(z_t) \quad (1.1)$$

Let A be a compact set that is invariant under Φ , so $\Phi(A) = A$, and that contains the initial state z_0 . We assume that Φ is a diffeomorphism on an open neighbourhood of A . $\Phi(A) = A$ and Φ is injective, so z_t lies in A at every t .

The state z_t usually cannot be obtained directly. Instead, d variables are observed at the times $t = 0, 1, \dots, T-1$, spaced at a fixed interval. Variable $i \in \{1, \dots, d\}$ has an **observation function** g_i , smooth on that neighbourhood, and its **observation** at time t is

$$x_t^{(i)} = g_i(z_t) \quad (1.2)$$

We write $x_t = (x_t^{(1)}, \dots, x_t^{(d)}) \in \mathbb{R}^d$ for the observations at time t , and $x = (x_0, \dots, x_{T-1})$ for the observations at all times.

2. Delay coordinates

We represent the state z_t by observations at selected variables and lags. We call the ordered sequence of selected (variable index, lag) pairs the **delay coordinates**,

$$\mathcal{J} = ((i_1, \ell_1), \dots, (i_E, \ell_E)), \quad i_j \in \{1, \dots, d\}, \quad \ell_j \in \mathbb{Z} \quad (2.1)$$

Each pair (i_j, ℓ_j) is one delay coordinate. The number $E = |\mathcal{J}|$ is the dimension of the resulting delay vector. Evaluating x at the selected delay coordinates gives the **delay vector** at time t ,

$$u_{\mathcal{J}}(t) = (x_{t-\ell_1}^{(i_1)}, \dots, x_{t-\ell_E}^{(i_E)}) \in \mathbb{R}^E \quad (2.2)$$

and its domain is the set of times at which all coordinates are defined,

$$\text{dom } u_{\mathcal{J}} = \{t \in \mathbb{Z} : 0 \leq t - \ell_j \leq T-1 \text{ for all } j\} = \left\{t \in \mathbb{Z} : \max_j \ell_j \leq t \leq T-1 + \min_j \ell_j\right\} \quad (2.3)$$

Taking one variable at evenly spaced lags gives the classical univariate delay coordinates,

$$\mathcal{J} = ((1, 0), (1, \tau), \dots, (1, (E-1)\tau)) \quad (2.4)$$

with lag τ . Allowing i_j to vary gives multivariate delay coordinates. A negative lag ℓ_j selects the observation of variable i_j taken $|\ell_j|$ steps after time t .

Every entry of the delay vector $x_{t-\ell_j}^{(i_j)}$ can be determined by the state at the single time t . It is the observation of variable i_j at time $t - \ell_j$,

$$x_{t-\ell_j}^{(i_j)} = g_{i_j}(z_{t-\ell_j}) \quad (2.5)$$

and invertibility (Φ is a diffeomorphism) makes the forward or inverse iterate of Φ well defined,

$$z_{t-\ell_j} = \Phi^{-\ell_j}(z_t) \quad (2.6)$$

Composing the observation function with the dynamics, the entry can be written as a function of z_t ,

$$x_{t-\ell_j}^{(i_j)} = g_{i_j}(\Phi^{-\ell_j}(z_t)) \quad (2.7)$$

Collecting these entries as functions of the state defines the **delay-coordinate map** $\Psi_{\mathcal{J}}$,

$$\Psi_{\mathcal{J}}(z) = (g_{i_1}(\Phi^{-\ell_1}(z)), \dots, g_{i_E}(\Phi^{-\ell_E}(z))) \quad (2.8)$$

The delay vector is the image of the current state under this map,

$$u_{\mathcal{J}}(t) = \Psi_{\mathcal{J}}(z_t) \quad (2.9)$$

3. The function from $u_{\mathcal{J}}(t)$ to $u_{\mathcal{K}}(t)$

We introduce a function Γ from $u_{\mathcal{J}}(t)$ to $u_{\mathcal{K}}(t)$, show that Γ exists and is continuous under a condition on $\mathcal{J}$ and $\mathcal{K}$, and estimate Γ from the observations.

The states z_t , the dynamics Φ and the observation functions g_i are unknown, and we want to learn about them from the observations. Take two sets of delay coordinates, $\mathcal{J}$ of dimension E and $\mathcal{K}$ of dimension E' , with delay-coordinate maps

$$\Psi_{\mathcal{J}} : A \rightarrow \mathbb{R}^E, \quad \Psi_{\mathcal{K}} : A \rightarrow \mathbb{R}^{E'} \quad (3.1)$$

We learn about z_t , Φ and g_i through a function $\Gamma : \Psi_{\mathcal{J}}(A) \rightarrow \mathbb{R}^{E'}$,

$$u_{\mathcal{K}}(t) = \Gamma(u_{\mathcal{J}}(t)) \quad (3.2)$$

We want to learn about the true Φ and g_i , which are defined on all of A , so we require this at every state of A :

$$\Gamma(\Psi_{\mathcal{J}}(z)) = \Psi_{\mathcal{K}}(z), \quad z \in A \quad (3.3)$$

If such a Γ exists, then

$$\Psi_{\mathcal{J}}(z) = \Psi_{\mathcal{J}}(z') \Rightarrow \Psi_{\mathcal{K}}(z) = \Psi_{\mathcal{K}}(z'), \quad z, z' \in A \quad (3.4)$$

Conversely, if Eq. (3.4) holds, then $\Gamma(\Psi_{\mathcal{J}}(z)) := \Psi_{\mathcal{K}}(z)$ is well defined and satisfies Eq. (3.3). So a Γ satisfying Eq. (3.3) exists if and only if Eq. (3.4) holds. We want Eq. (3.3), so we assume Eq. (3.4) for $\mathcal{J}$ and $\mathcal{K}$.

$\Psi_{\mathcal{J}}$ and $\Psi_{\mathcal{K}}$ are defined by Φ and the g_i , which are unknown, so Γ cannot be computed from Eq. (3.3). The observations are all that we have, so we want to obtain Γ from them. The observations give the pairs $(u_{\mathcal{J}}(t), u_{\mathcal{K}}(t))$ at finitely many t , and each pair satisfies $u_{\mathcal{K}}(t) = \Gamma(u_{\mathcal{J}}(t))$, so the observations give the values of Γ at finitely many points of $\Psi_{\mathcal{J}}(A)$. The values of a function at finitely many points do not determine its values at other points, so we want a property of Γ that relates its values at different points. Φ and the g_i are smooth, so $\Psi_{\mathcal{J}}$ and $\Psi_{\mathcal{K}}$ are continuous. A is compact and $\Psi_{\mathcal{J}}(A) \subset \mathbb{R}^E$ is Hausdorff, so the continuous surjection $\Psi_{\mathcal{J}} : A \rightarrow \Psi_{\mathcal{J}}(A)$ is a closed map and therefore a quotient map¹. A map on the image of a quotient map is continuous if and only if its composition with the quotient map is continuous, and $\Gamma \circ \Psi_{\mathcal{J}} = \Psi_{\mathcal{K}}$ is continuous, so Γ is continuous on $\Psi_{\mathcal{J}}(A)$. Γ is continuous at each q of $\Psi_{\mathcal{J}}(A)$, so $\Gamma(u_{\mathcal{J}}(t))$ lies within any given distance of $\Gamma(q)$ whenever $u_{\mathcal{J}}(t)$ lies close enough to q . Each $u_{\mathcal{K}}(t)$ equals $\Gamma(u_{\mathcal{J}}(t))$, so the pairs whose $u_{\mathcal{J}}(t)$ lies close enough to q give $u_{\mathcal{K}}(t)$ within that distance of $\Gamma(q)$. We want Γ at every q of $\Psi_{\mathcal{J}}(A)$, so we estimate $\Gamma(q)$ from the pairs whose $u_{\mathcal{J}}(t)$ lies near q .

Eq. (3.4) is a condition on $\Psi_{\mathcal{J}}$ and $\Psi_{\mathcal{K}}$, which are unknown, so it cannot be checked from the observations. We want Eq. (3.4) to hold for every $\mathcal{K}$ we may choose, so we want an assumption on $\mathcal{J}$ alone that implies Eq. (3.4) for every $\mathcal{K}$. $\Psi_{\mathcal{J}}$ is **injective** on A if it takes distinct values at distinct states,

$$\Psi_{\mathcal{J}}(z) = \Psi_{\mathcal{J}}(z') \Rightarrow z = z', \quad z, z' \in A \quad (3.5)$$

¹Munkres (2000), *Topology*, 2nd ed., Prentice Hall, §22 and §26: a continuous map from a compact space to a Hausdorff space is closed, a closed continuous surjection is a quotient map, and a map out of a quotient space is continuous exactly when its composition with the quotient map is continuous.

If $\Psi_{\mathcal{J}}$ is injective, then $\Psi_{\mathcal{J}}(z) = \Psi_{\mathcal{J}}(z')$ gives $z = z'$ and hence $\Psi_{\mathcal{K}}(z) = \Psi_{\mathcal{K}}(z')$, so Eq. (3.4) holds for every $\mathcal{K}$. An injective $\Psi_{\mathcal{J}}$ is a bijection from A onto $\Psi_{\mathcal{J}}(A)$, so its inverse $\Psi_{\mathcal{J}}^{-1} : \Psi_{\mathcal{J}}(A) \rightarrow A$ is defined and $z_t = \Psi_{\mathcal{J}}^{-1}(u_{\mathcal{J}}(t))$. Substituting $z = \Psi_{\mathcal{J}}^{-1}(q)$ into Eq. (3.3) gives Γ for every $\mathcal{K}$,

$$\Gamma(q) = (\Psi_{\mathcal{K}} \circ \Psi_{\mathcal{J}}^{-1})(q), \quad q \in \Psi_{\mathcal{J}}(A) \quad (3.6)$$

$\Psi_{\mathcal{J}}$ is a quotient map, so when it is injective its inverse is continuous, and $\Psi_{\mathcal{J}}$ is an **embedding** of A into $\mathbb{R}^E$. Whether $\Psi_{\mathcal{J}}$ is injective depends on Φ and the g_i , so injectivity cannot be checked from the observations either. Embedding theorems give conditions on $\mathcal{J}$ under which $\Psi_{\mathcal{J}}$ is an embedding for all Φ and g_i outside an exceptional set: for delay coordinates with several variables and arbitrary lags, it suffices that E exceeds twice the box-counting dimension of A and that Φ satisfies conditions on its periodic orbits². Φ is a diffeomorphism on an open neighbourhood of A and the g_i are smooth there, as these theorems require. We take E larger than twice the box-counting dimension of A and assume the conditions on periodic orbits, so $\Psi_{\mathcal{J}}$ is an embedding unless Φ and the g_i lie in the exceptional set.

4. The regression function

Sections 1 and 3 assumed deterministic dynamics, observations without measurement noise and Eq. (3.4), and under these assumptions we estimate Γ . We define the function F that we estimate in general, the conditional mean of $u_{\mathcal{K}}(t)$ given $u_{\mathcal{J}}(t)$, and state the assumptions under which F is determined by the joint distribution of the pair $(u_{\mathcal{J}}(t), u_{\mathcal{K}}(t))$ and is continuous. Under the assumptions of Sections 1 and 3, $F = \Gamma$.

Each of the three assumptions can fail. Φ and the g_i may lie in the exceptional set of the embedding theorems, so that $\Psi_{\mathcal{J}}$ is not injective. Then Eq. (3.4) can fail: two states with the same $\Psi_{\mathcal{J}}$ have different $\Psi_{\mathcal{K}}$. The observations may contain measurement noise, so that $u_{\mathcal{J}}(t)$ and $u_{\mathcal{K}}(t)$ are $\Psi_{\mathcal{J}}(z_t)$ and $\Psi_{\mathcal{K}}(z_t)$ plus errors that vary with t . The dynamics may be stochastic, so that z_{t+1} is not a function of z_t . In each case one value q of $u_{\mathcal{J}}(t)$ can occur with several values of $u_{\mathcal{K}}(t)$, so no function Γ satisfies $u_{\mathcal{K}}(t) = \Gamma(u_{\mathcal{J}}(t))$ at every t .

We want to estimate a value at q from the pairs in these cases too, so we want a function F that assigns one value to each q . A value computed from the pairs depends on T and on the initial state z_0 , so we want F to be defined by a probability distribution of the pair $(u_{\mathcal{J}}(t), u_{\mathcal{K}}(t))$ and not by the observations. Under deterministic dynamics the pair is the function $(\Psi_{\mathcal{J}}, \Psi_{\mathcal{K}})$ of z_t , so a distribution of the pair is given by a distribution of z_0 . With measurement noise or stochastic dynamics, it is given by the probability law of the noise or of the dynamics. This distribution can depend on t . We want one F for every t , so we assume that the joint distribution of the pair is the same at every t . Under deterministic dynamics, let μ be a probability measure on A that is invariant under Φ , and take z_0 distributed according to μ . Invariance makes the distribution of z_t equal to μ at every t , so the joint distribution of the pair is the same at every t .

We want $F(q)$ to be the single value at the smallest mean squared distance from the values of $u_{\mathcal{K}}(t)$ that occur with q . Among all measurable functions of $u_{\mathcal{J}}(t)$, the conditional mean of $u_{\mathcal{K}}(t)$ given $u_{\mathcal{J}}(t)$ uniquely minimizes the mean squared error, up to values on sets of probability zero³, so we define F as this conditional mean, the **regression function**,

²Takens (1981), “Detecting strange attractors in turbulence,” *Dynamical Systems and Turbulence, Warwick 1980*, Lecture Notes in Mathematics 898:366–381, proves the result for $2m + 1$ consecutive lags of one variable on a compact m -dimensional manifold, with the exceptional set meagre (the conclusion holds for *generic* Φ and g). Sauer, Yorke & Casdagli (1991), “Embedology,” *Journal of Statistical Physics* 65(3–4):579–616, replace m by the box-counting dimension D of a compact invariant set and require $E > 2D$, with the exceptional set of measure zero in the sense of *prevalence*. Deyle & Sugihara (2011), “Generalized theorems for nonlinear state space reconstruction,” *PLoS ONE* 6(3):e18295, extend the result to several observation functions and non-consecutive lags. Barański, Gutman & Śpiewak (2020), “A probabilistic Takens theorem,” *Nonlinearity* 33(9):4940–4966, fix an invariant probability measure and obtain injectivity on a set of full measure when E exceeds the Hausdorff dimension of that measure.

³Györfi, Kohler, Krzyżak & Walk (2002), *A Distribution-Free Theory of Nonparametric Regression*, Springer Series in Statistics, chapter 1.

$$F(q) = \mathbb{E}[u_{\mathcal{K}}(t) \mid u_{\mathcal{J}}(t) = q] \quad (4.1)$$

Its domain is the support of the distribution of $u_{\mathcal{J}}(t)$, and each value $F(q)$ lies in $\mathbb{R}^{E'}$. F is defined by the joint distribution of the pair, which is the same at every t , so F depends neither on t nor on the observations.

The observations are taken along one trajectory of the state, and we want to estimate F from their pairs, so we want the fraction of times at which the pair lies in a set to converge to the probability of that set as T grows. Under deterministic dynamics this holds for each measurable set and μ -almost every z_0 when μ is ergodic⁵, so we assume that μ is ergodic. With measurement noise or stochastic dynamics, we assume the same convergence for the sequence of pairs.

Under deterministic dynamics, observations without measurement noise and Eq. (3.4), $u_{\mathcal{K}}(t) = \Gamma(u_{\mathcal{J}}(t))$ at every t , so the conditional distribution of $u_{\mathcal{K}}(t)$ given $u_{\mathcal{J}}(t) = q$ is concentrated at $\Gamma(q)$ and $F = \Gamma$ on the domain of F . Γ is continuous on $\Psi_{\mathcal{J}}(A)$ and the domain of F lies in $\Psi_{\mathcal{J}}(A)$, so F is continuous. We estimate $F(q)$ from the pairs whose $u_{\mathcal{J}}(t)$ lies near q , as we estimate $\Gamma(q)$, so we want F to be continuous in the other cases too. In the other cases F need not be continuous, and we assume that it is.

5. 5. The three matrices: X , Y and Q

Estimating F from the observations requires two sequences of times. In this estimation problem, $u_{\mathcal{J}}$ serves as an input delay vector and its paired $u_{\mathcal{K}}$ as an output delay vector. The **library times** $L = (t_1, \dots, t_N)$ are times at which both $u_{\mathcal{J}}$ and $u_{\mathcal{K}}$ can be read. The **query times** $S = (s_1, \dots, s_M)$ are times at which $u_{\mathcal{J}}$ can be read and $u_{\mathcal{K}}$ is to be estimated.

Library times require both delay vectors, while query times require only the input delay vector.

$$L \subseteq \text{dom } u_{\mathcal{J}} \cap \text{dom } u_{\mathcal{K}}, \quad S \subseteq \text{dom } u_{\mathcal{J}} \quad (5.1)$$

For any sequence of valid times $(r_1, \dots, r_n)$, stack the corresponding delay vectors into the matrix

$$U_{\mathcal{J}}((r_1, \dots, r_n)) = \begin{pmatrix} u_{\mathcal{J}}(r_1) \\ \vdots \\ u_{\mathcal{J}}(r_n) \end{pmatrix} \in \mathbb{R}^{n \times E} \quad (5.2)$$

For example, let $d = 1$ and $x_t = t$ for $t = 0, 1, \dots, 9$, and take

$$\mathcal{J} = ((1, 0), (1, 2), (1, 4)) \quad (5.3)$$

Then $\text{dom } u_{\mathcal{J}} = \{4, 5, \dots, 9\}$ and

$$U_{\mathcal{J}}((4, 5, \dots, 9)) = \begin{pmatrix} 4 & 2 & 0 \\ 5 & 3 & 1 \\ 6 & 4 & 2 \\ 7 & 5 & 3 \\ 8 & 6 & 4 \\ 9 & 7 & 5 \end{pmatrix} \quad (5.4)$$

Applying $u_{\mathcal{J}}$ and $u_{\mathcal{K}}$ to these two sequences gives three matrices.

$$X = U_{\mathcal{J}}(L), \quad Y = U_{\mathcal{K}}(L), \quad Q = U_{\mathcal{J}}(S) \quad (5.5)$$

⁴For a continuously distributed $u_{\mathcal{J}}(t)$ the event $u_{\mathcal{J}}(t) = q$ has probability zero, and the conditional mean is defined through a regular conditional distribution of $u_{\mathcal{K}}(t)$ given $u_{\mathcal{J}}(t)$.

⁵Walters (1982), *An Introduction to Ergodic Theory*, Graduate Texts in Mathematics 79, Springer, states the invariant-measure and ergodic framework used here.

Each row corresponds to one time and each column to one delay coordinate. X contains input delay vectors at the library times, Y contains output delay vectors at those same times, and Q contains input delay vectors at the query times.

Matrix	Size	One row per	One column per
X	$N \times E$	library time	input coordinate
Y	$N \times E'$	the same library time	output coordinate
Q	$M \times E$	query time	input coordinate

The four objects $\mathcal{J}$, $\mathcal{K}$, L and S encode four parts of the estimation problem.

- $\mathcal{J}$ selects the variables and lags used as input coordinates.
- $\mathcal{K}$ selects the variables and lags used as output coordinates, including the forecast horizon.
- L selects the pairs used for fitting.
- S selects the input delay vectors at which predictions are requested.

From here on, q denotes a single query point, one row of Q , and everything is written for that one point. The notation applies row by row even when several predictions share one fitted function or one set of tuning parameters.

6. 6. Estimating F from nearby pairs

Each library row is one pair (X_n, Y_n) , and Y_n is drawn from the conditional distribution given X_n , whose mean is $F(X_n)$. It can be written as

$$Y_n = F(X_n) + \varepsilon_n, \quad \mathbb{E}[\varepsilon_n \mid X_n] = 0 \quad (6.1)$$

Continuity connects a query point to these pairs. When X_n is close to q , $F(X_n)$ is close to $F(q)$. The residual ε_n remains, so several pairs must be combined to reduce its effect.

Four decisions determine that combination. The distance rule determines which input delay vectors count as similar. The loss weights determine how strongly pairs at different distances affect the fit. The fitted function class determines which variation of F can be represented near the query. Regularization determines how strongly unsupported variation is suppressed.

Simplex projection and S-map express similarity as query-dependent loss weights. GP-EDM uses equal loss weights and expresses similarity through a kernel in the fitted function class. All three can be written as regularized weighted least squares,

$$\hat{f}_q = \arg \min_{f \in \mathcal{F}} \sum_{n=1}^N w_n(q) \|Y_n - f(X_n)\|^2 + \lambda \Omega(f), \quad \hat{F}(q) = \hat{f}_q(q) \quad (6.2)$$

Here X_n and Y_n are row n of X and Y . The symbol $\mathcal{F}$ denotes the fitted function class, $w_{n(q)}$ the loss weight, and $\lambda \Omega(f)$ the regularization penalty. Query-dependent weights produce a separate fitted function $\hat{f}_q$ for each q . Equal weights produce one fitted function that is evaluated at every query point.

Four choices go into this form.

- **The function class** $\mathcal{F}$ specifies the functions that may be fitted.
- **The distance rule** specifies how differences between input delay vectors enter the loss weights or the kernel.
- **The loss weights** $w_{n(q)}$ specify the contribution of each squared residual.
- **The penalty** $\lambda \Omega(f)$ controls complexity and stabilizes underdetermined fits.

Simplex projection takes $\mathcal{F}$ to be the constant maps, truncates the weights to the k nearest neighbours, and needs no penalty. S-map takes $\mathcal{F}$ to be the affine maps, gives weight to every library point, and carries a Tikhonov penalty. GP-EDM takes $\mathcal{F}$ to be a reproducing kernel Hilbert space, weights every library point equally, and carries the squared norm of that space as its penalty. Simplex projection represents F by a constant over a truncated neighbourhood. S-map represents F by an affine function and changes the effective neighbourhood continuously. GP-EDM fits one function in a reproducing kernel Hilbert space and lets the kernel describe which inputs should have similar function values.

6.1. The common linear-smoother form

For the three choices just described, the objective is quadratic in the fitted coefficients. With positive simplex weights, positive S-map regularization and positive GP noise variance, the corresponding quadratic has a unique minimizer. After all tuning parameters are fixed, the prediction is linear in the library outputs,

$$\hat{F}(q) = \sum_{n=1}^N b_n(q) Y_n \quad (6.3)$$

where the coefficients $b_{n(q)}$ depend on X , q and the fixed estimator parameters. Write $b(q) \in \mathbb{R}^N$ for the vector they form, so that $\hat{F}(q) = b(q)^\top Y$. Stacking $b(q)^\top$ over the M query points gives an $M \times N$ matrix B with $\hat{Y} = BY$, called the **smoother matrix**; the three methods differ only in its rows.

- **Additional outputs reuse the smoother.** When the parameters are fixed and shared across output coordinates, adding columns to Y applies the same B to each column.
- **Changing library times requires a new smoother.** Removing paired rows from X and Y leaves $\mathcal{J}$, $\mathcal{K}$ and the estimator family unchanged, but the coefficients must be recomputed from the remaining rows.
- **The signs and sum of a row determine extrapolation.** The prediction lies in the convex hull of the Y_n for every Y exactly when the entries of $b(q)$ are nonnegative and sum to one. A row that only sums to one reproduces constant outputs but may extrapolate beyond their convex hull.

One qualification. B is independent of Y only after the parameters governing w , $\mathcal{F}$ and λ have been fixed. Selecting k , θ or kernel parameters from Y makes the complete fitting procedure nonlinear in Y . Prediction error used for parameter selection is also optimistic for that selected configuration unless evaluation uses pairs excluded from selection.

7. Simplex projection

Simplex projection is the case in which $\mathcal{F}$ is the constant maps and the weights are cut off beyond the k nearest neighbours. It treats F as constant across the selected neighbourhood, so its smoothing bias grows when F varies appreciably within that neighbourhood⁶.

Write $\delta_n = \|q - X_n\|$ for the Euclidean distance from the query point q to each library point, take the k smallest, and write $n_1, \dots, n_k$ for their indices and $\delta_{(1)} \leq \dots \leq \delta_{(k)}$ for the distances. The conventional choice $k = E + 1$ equals the number of vertices of an E -dimensional simplex. The selected neighbours need not enclose q , so the name describes their number rather than a geometric containment guarantee.

The weights are exponential in the distance, normalized by the distance to the nearest neighbour,

$$w_{n_a} = \exp\left(-\frac{\delta_{(a)}}{\delta_{(1)}}\right), \quad a = 1, \dots, k \quad (7.1)$$

and $w_n = 0$ at every library point outside the k nearest. This cut-off reduces the sum over all N library points to a sum over k of them. The displayed formula assumes $\delta_{(1)} > 0$. Its limit as $\delta_{(1)}$ approaches zero

⁶Sugihara & May (1990), “Nonlinear forecasting as a way of distinguishing chaos from measurement error in time series,” *Nature* 344(6268):734–741, introduced nonlinear forecasting with nearest neighbours to ecology.

assigns weight only to neighbours at zero distance; an implementation must state how it handles exact duplicates.

This normalization does two things. The weights become invariant to the overall scale of the distances, and the bandwidth narrows automatically where neighbours are dense. The nearest neighbour always takes weight e^{-1} , so the distance from the query point to its neighbourhood enters only through the ratios $\delta_{(a)}/\delta_{(1)}$.

Since the function class contains only constant maps, the minimization reduces to a weighted mean and no penalty is needed. The selected weights have a positive sum, so the minimizer is unique.

$$\hat{F}(q) = \frac{\sum_{a=1}^k w_{n_a} Y_{n_a}}{\sum_{a=1}^k w_{n_a}} \quad (7.2)$$

So $b_{n_a}(q) = w_{n_a} / \sum_c w_{n_c}$ at the k nearest library points, and $b_n(q) = 0$ at all the others. They are nonnegative and sum to one, so the prediction lies in the convex hull of the neighbouring output vectors.

8. 8. S-map

S-map is the case in which $\mathcal{F}$ is the affine maps and every library point is given a nonzero weight. An affine fit represents the first-order variation of a smooth regression function within the effective neighbourhood and thereby reduces the leading bias of a local constant fit⁷.

Write $\bar{\delta}(q) = \frac{1}{N} \sum_n \delta_n$ for the mean distance seen from the query point q and set

$$w_n = \exp\left(-\theta \frac{\delta_n}{\bar{\delta}(q)}\right) \quad (8.1)$$

At $\theta = 0$ every weight is 1 and the fit reduces to a single linear regression over the whole library. Raising θ concentrates the weight on the nearest neighbours and makes the fit local. The parameter θ tunes the bandwidth continuously, in place of the discrete cut-off at k .

The function class is the affine maps $f(p) = c_0 + C_1^\top p$. To absorb the intercept, extend the design matrix to $\tilde{X} = [\mathbf{1}, X] \in \mathbb{R}^{N \times (E+1)}$ and the query point to $\tilde{q} = (1, q) \in \mathbb{R}^{E+1}$, so that the coefficient matrix $C \in \mathbb{R}^{(E+1) \times E'}$ carries $c_0^\top$ on its first row and C_1 on the remaining E .

Redundant delay coordinates can make the columns of $\tilde{X}$ linearly dependent or nearly dependent. A large θ can create the same numerical problem by leaving only a few library points with appreciable weight. Autocorrelation often makes lagged coordinates similar, but it does not make them exactly collinear by definition.

The penalty is taken to be Tikhonov on the linear part, $\Omega(f) = \|C_1\|_F^2$, and with $W = \text{diag}(w_1, \dots, w_N)$ the normal equations read $GC = \tilde{X}^\top W Y$ with

$$G = \tilde{X}^\top W \tilde{X} + \alpha \text{tr}(\tilde{X}^\top W \tilde{X}) I_0 \quad (8.2)$$

where I_0 is the identity with its $(1, 1)$ entry set to 0, so that the intercept is left unpenalized. Multiplying the penalty by the trace makes α a dimensionless multiplier of the local normal matrix. It does not make the fit invariant to rescaling one input coordinate relative to another. The solution gives the prediction $\hat{F}(q) = \tilde{q}^\top C$, and $b(q)^\top = \tilde{q}^\top G^{-1} \tilde{X}^\top W$.

G is positive definite whenever $\alpha > 0$ and some weight is nonzero, since

$$v^\top G v = \|W^{1/2} \tilde{X} v\|^2 + \alpha \text{tr}(\tilde{X}^\top W \tilde{X}) v^\top I_0 v \quad (8.3)$$

⁷Sugihara (1994), “Nonlinear forecasting for the classification of natural time series,” *Philosophical Transactions of the Royal Society of London A* 348(1688):477–495.

and the two vanish together only at $v = 0$: the second term forces $v_2 = \dots = v_{E+1} = 0$, and the first then reads $v_1^2 \sum_n w_n$.

Every library point has a nonzero weight, so no entry of that row is structurally zero, and the entries can be negative: the prediction is not confined to the convex hull of Y . Leaving the intercept unpenalized does keep the row summing to one, since a constant Y is then fitted with zero residual and zero penalty, so a constant library is still reproduced exactly. The weights depend on the query point and so does C , so one system is solved per query point.

The fitted slopes $C_1^\top$ are coefficients of the locally weighted affine approximation. They estimate the Jacobian $F'(q)$ only in an asymptotic regime where the effective neighbourhood shrinks, the local sample size grows, and F has the required derivatives⁸. When the input delay-coordinate map embeds A in the deterministic model, this Jacobian is the derivative of $\Psi_\kappa \circ \Psi_{\mathcal{J}}^{-1}$. With non-injective, noisy or stochastic input coordinates, it is the derivative of the conditional mean. A finite-sample S-map coefficient is therefore an estimate of a local predictive slope, not automatically a structural interaction coefficient.

An improvement in out-of-sample prediction at $\theta > 0$ shows that state-dependent weighting predicts better than the global affine fit on that evaluation distribution. This comparison supports state dependence of the predictive relationship only after alternatives such as nonstationarity, leakage and parameter-selection bias have been controlled. The pairs used to select θ cannot also provide an unbiased estimate of the selected model's improvement.

9. 9. GP-EDM

GP-EDM, empirical dynamic modelling with a Gaussian process, uses equal loss weights and a reproducing kernel Hilbert space as its function class⁹. The kernel encodes how rapidly function values may change with separation in each input coordinate. Its scale parameters are estimated from X and Y or assigned prior distributions rather than fixed by a local polynomial degree.

Fix a positive definite kernel κ on $\mathbb{R}^E$, write $\mathcal{H}_\kappa$ for its reproducing kernel Hilbert space, and take

$$w_n \equiv 1, \quad \mathcal{F} = \mathcal{H}_\kappa, \quad \Omega(f) = \|f\|_{\mathcal{H}_\kappa}^2, \quad \lambda = \sigma^2 \quad (9.1)$$

For one output coordinate, the minimization has a closed solution. Write $K \in \mathbb{R}^{N \times N}$ for the matrix with entries $\kappa(X_m, X_n)$ and $\kappa(q) \in \mathbb{R}^N$ for the vector with entries $\kappa(q, X_n)$.

$$\hat{F}(q) = \kappa(q)^\top (K + \sigma^2 I)^{-1} Y \quad (9.2)$$

Here $b(q)^\top = \kappa(q)^\top (K + \sigma^2 I)^{-1}$. The inverse exists because K is positive semidefinite and $\sigma^2 > 0$, which makes $K + \sigma^2 I$ positive definite. The equal weights are query-independent, so one function is fitted and then evaluated at all M query points, rather than the M separate fits the other two methods require. The prediction is still linear in Y with query-dependent coefficients, and the inverse is computed once and reused across the whole query.

The same expression is the posterior mean of a Gaussian process prior $f \sim \mathcal{GP}(0, \kappa)$ observed with independent Gaussian noise of variance σ^2 . The model also gives the posterior predictive variance for a new noisy output,

$$\hat{\sigma}^2(q) = \kappa(q, q) - \kappa(q)^\top (K + \sigma^2 I)^{-1} \kappa(q) + \sigma^2 \quad (9.3)$$

⁸Masry (1996), “Multivariate local polynomial regression for time series: uniform strong consistency and rates,” *Journal of Time Series Analysis* 17(6):571–599, proves derivative consistency under stationarity, smoothness, bandwidth and strong-mixing conditions.

⁹Munch, Poynor & Arriaza (2017), “Circumventing structural uncertainty: a Bayesian perspective on nonlinear forecasting for ecology,” *Ecological Complexity* 32:134–143. Munch, Rogers & Sugihara (2023), “Recent developments in empirical dynamic modelling,” *Methods in Ecology and Evolution* 14(3):732–745. Rasmussen & Williams (2006), *Gaussian Processes for Machine Learning*, MIT Press, chapter 2, gives the Gaussian-process identities used here.

This quantity is calibrated as conditional uncertainty only when the Gaussian process covariance, mean and noise model are adequate. It is model-based uncertainty, whereas the regression function F is defined without a Gaussian process assumption.

9.1. Kernel similarity

Equal loss weights make GP-EDM a single global fit. Local similarity enters through the squared-exponential kernel, with one inverse squared length-scale parameter per input coordinate,

$$\kappa(p, p') = \eta \exp \left(- \sum_{j=1}^E \varphi_j (p_j - p'_j)^2 \right), \quad \varphi_j \geq 0 \quad (9.4)$$

The direct covariance $\kappa(q, X_n)$ decreases as the scaled separation from q increases. The final smoother coefficient $b_{n(q)}$ also contains $(K + \sigma^2 I)^{-1}$, so it need not decrease monotonically with distance and can be negative. The φ_j play the bandwidth role inside the function class that k and θ play inside the loss weights.

Two things change with the move.

The kernel scale is not renormalized separately at each query point. Simplex projection divides distances by the nearest-neighbour distance, while S-map divides them by the mean distance from the current query. A stationary GP kernel uses the same φ_j at every query point.

The decay is anisotropic. A separate φ_j per coordinate defines a fitted diagonal squared-distance rule in place of an unscaled Euclidean distance. This parameterization is called **automatic relevance determination** (ARD). When φ_j is near zero, changing coordinate j has little effect on the fitted covariance and the coordinate has little predictive relevance for that output. ARD does not show that the remaining coordinates embed the state; predictive relevance for a specified output and state reconstruction are different properties.

Rows of B are generally dense and signed, as in S-map, and they are not constrained to sum to one. The σ^2 in the inverse shrinks the prediction toward the prior mean, and by more where the library is sparse. Consequently, exact reproduction of a constant Y is not guaranteed, and the prediction is not confined to the convex hull of the library outputs.

9.2. Fitting the kernel

η , the φ_j and σ^2 can be chosen by maximizing the marginal likelihood of the library. For one output coordinate and a zero prior mean, the log marginal likelihood is

$$\log p(Y | X) = -\frac{1}{2} Y^\top (K + \sigma^2 I)^{-1} Y - \frac{1}{2} \log \det(K + \sigma^2 I) - \frac{N}{2} \log 2\pi \quad (9.5)$$

The quadratic term measures agreement with Y under the covariance model. The log-determinant term accounts for the volume of output vectors to which that covariance assigns substantial density. Together they balance agreement with Y against covariance flexibility. A prior on the hyperparameters changes the criterion from maximum likelihood to a posterior mode; the GP-EDM formulation of Munch, Poynor and Arriaza uses an ARD prior that favours negligible effects unless X and Y support them. With $E' > 1$ this is one Gaussian process per column of Y . Sharing the hyperparameters across the columns keeps a single smoother matrix, and with it the freedom to widen Y at no cost; fitting them per column gives one smoother matrix per column instead, and that freedom is lost.

The effect on input-coordinate relevance is the reason the fitting step matters here. One may supply a broad set of candidate variables and lags in $\mathcal{J}$ and use the fitted φ_j to identify coordinates that affect prediction of the chosen output. This procedure selects predictors for that regression function. Injectivity remains a separate property governed by the delay-coordinate map.

Hyperparameter fitting changes the linear-smoother property. Once η , φ and σ^2 are fitted from Y , the smoother matrix depends on Y , and the complete fitting procedure is not linear in Y . The same dependence

arises when k or θ is selected by predictive performance. A higher-dimensional hyperparameter search increases the need for regularization and evaluation on pairs not used for selection.

10. 10. The three methods compared

The table collects the statistical choices made by each method.

	Simplex projection	S-map	GP-EDM
Assumed shape of F near q	constant	affine	kernel-controlled smoothness
Function class $\mathcal{F}$	constant maps	affine maps	$\mathcal{H}_\kappa$
Input-coordinate distance	Euclidean	Euclidean	diagonal, fitted (φ)
Weights $w_n(q)$	$\exp(-\delta_{(a)}/\delta_{(1)})$	$\exp(-\theta \delta_n/\bar{\delta}(q))$	1
Support of the weights	the k nearest	all N	all N
Bandwidth	per query, from $\delta_{(1)}$	per query, from $\bar{\delta}(q)$	fixed across query points, from φ
Penalty	none	Tikhonov on C_1 , strength α	$\sigma^2 \ f\ _{\mathcal{H}_\kappa}^2$
Row of B	k nonzero, nonnegative, sums to 1	generally dense, signed, sums to 1	generally dense, signed, sum unconstrained
Prediction	in the convex hull of Y	unrestricted	shrunk toward the prior mean
Uncertainty from the fitted model	not supplied	not supplied	predictive variance
Solved per query	k neighbours and k weights	an $(E+1) \times (E+1)$ system	nothing; one $N \times N$ solve serves all

11. 11. Designing input and output coordinates

An analysis specifies $\mathcal{J}$ for the input and $\mathcal{K}$ for the output. X and Q use the same $\mathcal{J}$ because their rows are compared in one input-coordinate space. Y uses $\mathcal{K}$, whose variables, lags and dimension may differ from those of $\mathcal{J}$. The same pair of delay coordinates can be used with any of the three estimators.

The examples take two variables ($d = 2$). An output horizon of h is represented by an output lag of $-h$.

Goal	Input $\mathcal{J}(X, Q)$	Output $\mathcal{K}(Y)$	Sizes (X, Y)
One variable, horizon 0	$((1, 0), (1, \tau))$	$((1, 0))$	$N \times 2, N \times 1$
One variable to another, h ahead	$((1, 0), (1, \tau))$	$((2, -h))$	$N \times 2, N \times 1$
Several variables to one, h ahead	$((1, 0), (1, \tau), (2, 0), (2, \tau))$	$((1, -h))$	$N \times 4, N \times 1$
Several variables to several, h ahead	$((1, 0), (1, \tau), (2, 0), (2, \tau))$	$((1, -h), (2, -h))$	$N \times 4, N \times 2$
Different lags per variable	$((1, 0), (1, \tau), (2, 2\tau))$	$((1, -h))$	$N \times 3, N \times 1$
Several horizons	$((1, 0), (1, \tau))$	$((1, -h_1), (1, -h_2))$	$N \times 2, N \times 2$

Same variable at horizon 0. This output is degenerate because Y is the first column of X and is already contained in the input vector.

One variable to another. It asks whether the delay vectors built from variable 1 recover variable 2 at h steps ahead, so a forecast in time and a map between variables are carried out in a single operation. With $h = 0$ the row is **cross mapping**: the delay vectors built from one variable are asked for the contemporaneous value of another. **Convergent cross mapping** (CCM) repeats this estimate over increasing library sizes and evaluates how cross-map skill changes with $|L|$. Under the deterministic coupled-system assumptions of CCM, an effect variable contains information about the state of its causes. Consequently, using variable 1 delay coordinates to recover variable 2 tests the direction variable 2 $\rightarrow$ variable 1, not the reverse. If variable 1 reconstructs the shared state, increasing the library supplies closer analogues and the cross-map estimate of variable 2 improves before approaching its finite-noise limit. Convergence is therefore evidence consistent with that causal direction only when common forcing, synchrony, temporal trends, sampling effects and alternative directions have been addressed¹⁰.

Adding input coordinates. The number of library points is unchanged while the input-coordinate dimension grows from 2 to 4. The new coordinates may separate states that previously looked identical, but they also make close neighbours harder to find in a finite library unless the added coordinates are redundant on the sampled state set.

Adding output coordinates. The input distances and neighbours are unchanged, and only the number of columns of Y grows. Several variables can therefore reuse one fixed smoother matrix when estimator parameters are shared across output coordinates.

Different lags per variable. Since $\mathcal{J}$ is only a sequence of (variable, lag) pairs, neither the number of coordinates nor the spacing of the lags has to agree across variables. The displayed delay coordinates read variable 1 at lags 0 and τ and variable 2 at lag 2τ only. This specifies delay coordinates that mix variables and time scales. Generalized embedding theorems cover such delay coordinates when their regularity, dimension and periodic-orbit conditions hold.

Several output horizons. Each coordinate of $\mathcal{K}$ carries its own lag, so the horizons h_1 and h_2 are estimated at once. With fixed estimator parameters, neighbours and smoother coefficients are determined by the input alone, so another horizon adds another output column. Direct estimation does not feed earlier predictions into later inputs, so it avoids recursive propagation of earlier prediction errors.

One combination the table omits takes $\mathcal{K} = \mathcal{J}$ with every lag shifted by $-h$. The columns of Y then carry the input coordinates advanced by h steps, so the predicted row has the coordinates required for another application of the same fitted map. This alignment makes iterated multi-step forecasting possible.

Taking the query to be the library itself, $S = L$, and excluding each library point from its own neighbourhood gives leave-one-out prediction. A **Theiler window** widens the exclusion by removing every library time within r sampling steps of the query time¹¹. In prediction, this window prevents overlapping or strongly dependent delay vectors from serving as nominally independent validation cases. The required width depends on the lags, forecast horizon, serial dependence and intended deployment setting.

12. 12. Assumptions and limits

The regression function remains meaningful beyond deterministic dynamics, but each interpretation requires its own assumptions.

¹⁰Sugihara, May, Ye, Hsieh, Deyle, Fogarty & Munch (2012), “Detecting causality in complex ecosystems,” *Science* 338(6106):496–500.

¹¹Theiler (1986), “Spurious dimension from correlation algorithms applied to limited time-series data,” *Physical Review A* 34(3):2427–2432, introduced temporal exclusion to prevent serial dependence from creating spurious near neighbours in correlation-dimension estimation.

- **Stable relationship:** the pairs at library times and at query times must have the same conditional mean F . A regime change can alter F even when input delay vectors remain numerically close.
- **Support:** a local estimate is supported only where the library contains input delay vectors near the query. A query outside the sampled support requires extrapolation, regardless of the estimator's algebraic ability to return a number.
- **Dynamical interpretation:** interpreting F as $\Psi_{\mathcal{K}} \circ \Psi_{\mathcal{J}}^{-1}$ requires deterministic dynamics and an input delay-coordinate map that embeds A . Otherwise F is a conditional mean, and causal or structural language needs additional assumptions.
- **Temporal dependence:** ergodicity justifies long-run averages but supplies neither an effective sample size nor a convergence rate. Rates for local regression under temporal dependence require stronger conditions such as mixing. Validation splits and uncertainty calculations must preserve the temporal information available at the intended prediction time.
- **Leakage:** output times used for evaluation must be absent from fitting and parameter selection. Overlapping delay windows and temporally adjacent states can also make nominal train and test cases nearly identical. A Theiler window enforces separation when that separation matches the intended prediction task.
- **Coordinate scale:** Euclidean distance changes when one input coordinate is rescaled. Standardization, a physically chosen metric, or fitted kernel scales are modelling choices. ARD estimates relative predictive scales conditional on the units, kernel and hyperprior; it does not remove the need to state those choices.
- **Parameter selection:** choosing $\mathcal{J}$, $\mathcal{K}$, k , θ , α or kernel parameters from prediction performance uses output information. Performance of the selected configuration requires evaluation pairs that played no part in that selection.
- **Dimension:** finite libraries become sparse as the effective dimension of the input distribution grows. For independent pairs whose inputs lie on a smooth lower-dimensional manifold, local polynomial regression can attain rates governed by the manifold dimension rather than the ambient coordinate dimension¹². This theorem does not by itself establish an EDM rate on a fractal attractor with dependent pairs. Measurement noise can also spread the input delay vectors away from a lower-dimensional set, making the ambient dimension relevant in finite samples.
- **Skill measure:** Pearson correlation ρ is unchanged by a positive affine transformation of the predictions. It therefore does not detect additive bias or multiplicative miscalibration. Error and calibration measures are needed when their magnitudes matter.

The framework fixes the regression function, the three matrices and the estimator families. An analysis additionally fixes the input coordinates, output coordinates, library times, query times, distance scale, tuning procedure and evaluation measure.

¹²Bickel & Li (2007), "Local polynomial regression on unknown manifolds," *IMS Lecture Notes–Monograph Series* 54:177–186.